

1. Determine all critical points of $f(x) = \sqrt{1-x^2}$.
2. Consider the function $f(x) = \frac{1}{\sqrt{x-2}}$
 - a) Confirm that $x = 2$ either makes $f'(x)$ equal 0 or become undefined.
 - b) Explain why $x = 2$ is *NOT* a critical point of $f(x)$.
3. Find all critical points of $f(x) = \frac{x^2}{x^3+8}$.
4. Find all critical points of $y = x^2(x-1)^{2/3}$.
5. Find all critical points of $f(x) = \sin|x|$.
6. Consider the function $f(x) = \frac{ax+b}{x^2-1}$ where a and b are real numbers. Find values of a and b such that $f(x)$ has a local extreme value of 1 at $x = 3$.
7. TRUE or FALSE: A function $f(x)$ must have an absolute maximum or minimum on $[a, b]$.
8. TRUE or FALSE: The value $y = -10$ cannot be the absolute maximum of a function $f(x)$.
9. Find the absolute maximum and absolute minimum values of $f(x) = x^3 - 3x^2 - 9x + 1$ on $[-2, 1]$.
10. One way of proving that $f(x) \leq g(x)$ for all x in a given interval is to show that $0 \leq g(x) - f(x)$ for all x in the interval; and one way of proving the later inequality is to show that the absolute minimum of $g(x) - f(x)$ on the interval is non-negative. Use this idea to prove that $\sin(x) \leq x$ for all x in the interval $[0, 2\pi]$.
11. The concentration $C(t)$ of a drug in the bloodstream t hours after it has been injection is commonly model by the equation $C(t) = \frac{K(e^{-bt} - e^{-at})}{a-b}$, where $K > 0$ & $a > b > 0$. At what time does the maximum concentration occur?